Experiment 2 :Path Classification and Interaction Analysis

Objective: Identify the physical scene objects and interaction types associated with the propagation paths generated by Sionna RT at 5.9 GHz.

Method: The path solver's delay, validity, interaction-code, and object-ID arrays were inspected. Numeric object IDs were mapped to scene-object names.

Findings:

1. The direct path has a delay of approximately 133.426 ns, equivalent to 40.000 m of propagation.
2. A floor-reflected path was identified at approximately 134.112 ns, equivalent to 40.206 m.
3. Additional paths involve the scene objects `building_6` and `no-name-1`.
4. Several paths contain multiple reported specular-reflection interactions.
5. Two long-delay paths contain refraction interactions according to the solver metadata and require further verification.

Conclusion: The simulation produces a mixture of direct, floor-reflected, building-related, and refraction-labelled propagation paths. The results confirm that scene geometry and material interactions influence the simulated channel.

Limitation: Object `no-name-1` has not yet been identified geometrically. Refraction-labelled paths also require validation against the scene configuration before their physical interpretation is finalized.

Next task: Inspect the scene geometry and radio materials, identify `no-name-1`, and verify the refraction-labelled paths. Then proceed to a sampled channel impulse response and OFDM-based range estimation.

1. Path classification results

Path index	Delay (ns)	Equivalent distance (m)	Solver-reported interaction
0	133.426	40.000	Direct path; no interaction
9	134.112	40.206	Floor reflection
1	145.273	43.552	Building 6 reflection
2	147.918	44.345	`no-name-1` reflection
4, 5	180.333	54.063	Two building-related reflections
10, 11	180.842	54.215	Two building-related reflections and floor interaction
7	223.079	66.877	Three building-related interactions
8	228.234	68.423	Three building-related interactions
6	277.659	83.240	Refraction/reflection/refraction
3	283.282	84.926	Refraction/reflection/refraction

Note: The scene object `no-name-1` has not yet been identified by its geometry. We should not label it as a particular building until we inspect the scene.

2. What have learned

- Direct propagation: Path 0 matches the expected 40 m transmitter–receiver distance.
- Ground reflection: Path 9 is reported as a specular reflection from the floor, arriving only 0.686 ns after the direct path.
- Building multipath: Several paths involve `building_6` and `no-name-1`, illustrating how the street environment creates indirect propagation.
- Higher-order paths: Paths 7 and 8 contain three reported interactions, with longer propagation delays.
- Refraction-labelled paths: Paths 3 and 6 involve refraction according to the solver's interaction metadata. Their physical interpretation should be checked against the scene materials and geometry.